

SWIGGY INDIA



Data-Driven Investment Insights for the Indian Food Delivery Market

Sector: Food Technology / Restaurant Analytics

Tools: Python · Pandas · Tableau Public

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Context & Problem Statement

India's food delivery market — understanding the drivers of restaurant success

Market Context

India's food delivery sector has grown rapidly, driven by urbanisation, digital adoption, and shifting consumer lifestyles. Platforms like Swiggy have created highly competitive marketplaces where thousands of restaurants compete for visibility and ratings.

Customer ratings act as a key performance indicator — directly influencing discoverability, demand, and long-term success.

Problem Statement

Key analytical questions this project addresses:

1. Which cuisines and cities consistently achieve higher ratings?
2. Does pricing or discounting significantly impact customer satisfaction?
3. How do veg vs non-veg offerings influence performance?
4. Where do demand-supply mismatches exist in the market?
5. Which cuisine-city combinations represent the best entry opportunities?

Goal: Move beyond descriptive analysis → deliver decision-oriented insights that guide strategic positioning, investment decisions, and market entry in the Swiggy ecosystem.

Data Engineering

Source · Cleaning · Feature Engineering

10,083

Total Rows

7,738

Rated Restaurants

6

Cities Covered

80+

Cuisine Types

Data Cleaning

- Extracted numeric values from ratings; handled missing entries
- Standardised price column from text to numeric values
- Converted ratings count from abbreviated K-format to absolutes
- Removed corrupted / inconsistent fields
- Filtered records missing critical attributes

Feature Engineering

Price Band	Budget · Mid · Premium · Luxury segmentation
Rating Tier	Performance groupings for comparison
Demand Tier	Proxy using ratings count
Value Index	Rating-to-price ratio for value assessment
Primary Cuisine	Dominant cuisine extracted per listing

Source: <https://www.kaggle.com/datasets/rrkcoder/swiggy-restaurants-dataset/data>

KPI Framework

Baseline metrics defining overall market performance

3.996

Avg. Rating

Overall quality benchmark
across the platform

7,432

Total Restaurants

Effective market size
after filtering valid listings

₹305

Avg. Price (for 2)

Typical pricing level
across all restaurants

75.3%

% Rated

Platform maturity &
customer engagement

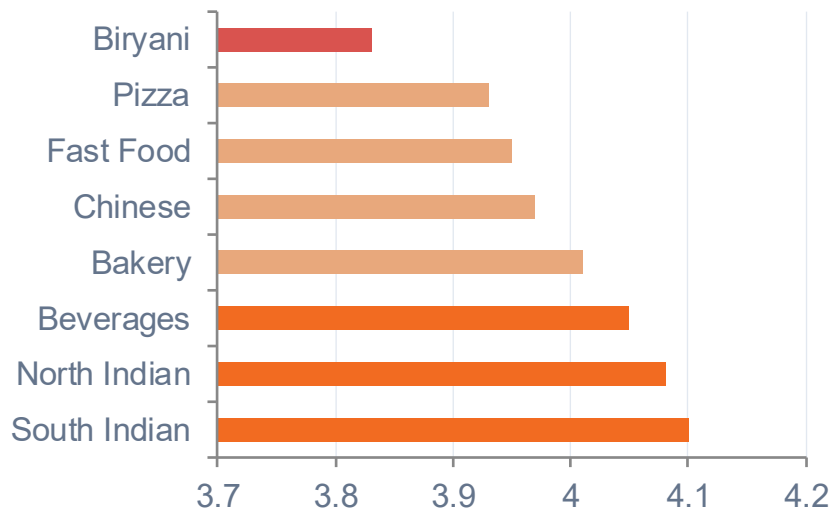
Business Interpretation

KPI	Value	Business Interpretation
Average Rating	3.996 (~4.00)	Represents the overall quality benchmark on the platform
Total Restaurants	7,432	Indicates effective market size after filtering valid listings
Average Price (for two)	₹305	Reflects the typical pricing level across all restaurants
% Rated Restaurants	75.3%	Measures platform maturity and customer engagement levels

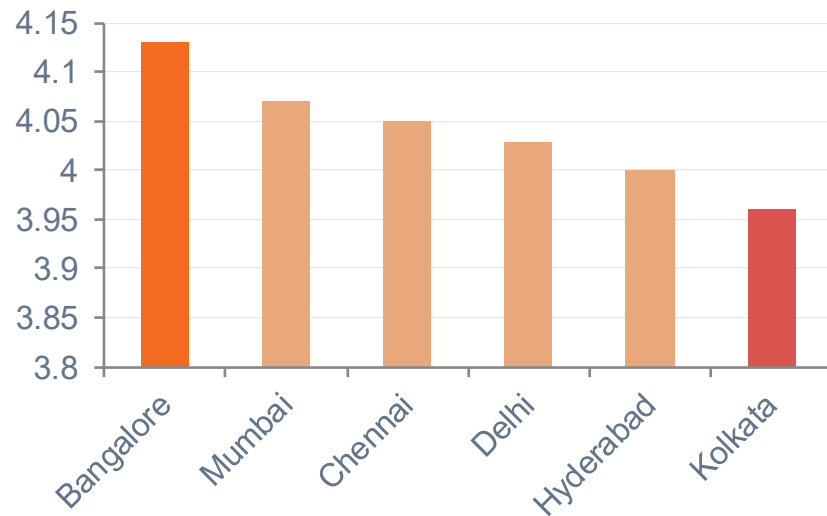
Key Insights from EDA

Exploratory data analysis across cuisine, city, and pricing dimensions

Rating by Cuisine



Rating by City



Key finding: Cuisine type and city-level dynamics are stronger drivers of performance than pricing — strategic positioning matters more than cost competition.

Advanced Statistical Analysis

Validating EDA findings with statistical techniques

1

Cuisine Impact on Ratings

Cuisine type is a statistically significant driver of ratings. High-performing categories — North Indian, Beverages, South Indian — consistently outperform. Differences across cuisine groups are significant.

2

Price vs Rating ($r \approx 0.1$)

Weak positive correlation between price and rating. Higher pricing does NOT guarantee higher ratings. Customer satisfaction depends more on value-for-money than absolute price.

3

Promotional Offers Impact

No significant relationship between number of offers and customer ratings. Discounts do not strongly influence perceived quality. Ratings reflect core experience — food quality and service.

4

Pure Veg vs Non-Veg

Pure vegetarian restaurants demonstrate slightly higher average ratings vs non-veg. Indicates consistent, reliable performance trend within the veg segment — strong consumer demand.

Dashboard Walkthrough

Three interactive Tableau dashboards — Executive · Operational · Market Insights

01

Dashboard

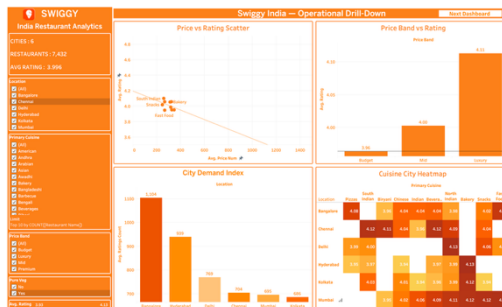
Executive Performance Dashboard



02

Dashboard

Operational Drill-Down Dashboard



03

Dashboard

Market Insights Dashboard



Tableau Public: https://public.tableau.com/app/profile/pratiti.paul/viz/SwiggyAnalysis_17773931032380/ExecutiveDashboard

Strategic Recommendations

5 actionable insights linked to analysis findings

01

Cuisine Strategy

Focus on High-Performing Cuisines

Prioritise South Indian, North Indian, and Beverages — consistently achieving higher ratings across cities. Increases probability of strong initial performance.

02

Location Strategy

Target High-Demand Urban Markets

Launch in Bangalore & Hyderabad where demand is highest. Stronger visibility, faster scaling, and higher customer engagement vs other cities.

03

Pricing Strategy

Adopt Mid-Premium Pricing

Position within Rs 300-500 price range — shows the highest average ratings. Optimal balance between perceived value and affordability.

04

Positioning

Leverage Pure Veg Positioning

Veg restaurants consistently outperform non-veg peers in ratings — a strong and stable demand segment with clear consumer preference.

05

Market Gaps

Exploit Demand-Quality Gaps

High-demand but lower-quality segments (e.g., Biryani in Delhi) offer high-risk, high-reward entry opportunities with superior quality.

Impact & Business Value

How these insights translate to tangible outcomes

4.1+

Target Rating

Revenue Optimisation

Aligning cuisine, pricing, and city targeting enables higher ratings, improved visibility, higher order volumes, and increased revenue potential.



Retention

Market Positioning

Focusing on high-performing cuisines and veg positioning drives differentiation, better customer retention, and stronger brand perception.



Risk

Expansion Efficiency

Data-driven location selection (Bangalore, Hyderabad) reduces failed expansion risk and minimises customer acquisition costs.

0

Discount Effect

Reduced Discount Dependency

Offers have minimal impact on ratings — restaurants can avoid unnecessary discounting and protect profit margins without affecting satisfaction.

Limitations & Future Scope

Acknowledging constraints and charting the path forward

Limitations

Snapshot in Time

Dataset represents a static view — does not capture seasonal or temporal trends.

Missing Ratings

~23% of restaurants lack ratings, potentially impacting accuracy of performance analysis.

No Revenue Data

Analysis cannot directly tie performance to profitability without order-level revenue data.

Platform-Specific

Insights are specific to Swiggy and may not generalise across other delivery platforms.

Future Scope

Time-Series Analysis

Incorporate temporal data to study trends, seasonality, and demand fluctuations.

Revenue Integration

Integrate revenue and order-level data to build predictive profitability models.

Tier-2 & Tier-3 Cities

Expand analysis to emerging markets to identify new growth opportunities.

Predictive ML Models

Develop pricing recommendation and rating prediction models using machine learning.

Thank You

Swiggy India Restaurant Analytics

DVA Capstone 2 · Group 19

Team Members

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